Programming Fundamentals Theory

Assignment

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Program: BCS-1K
Section: 1K
Course: Programming Fundamentals Theory
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Question No. 1:

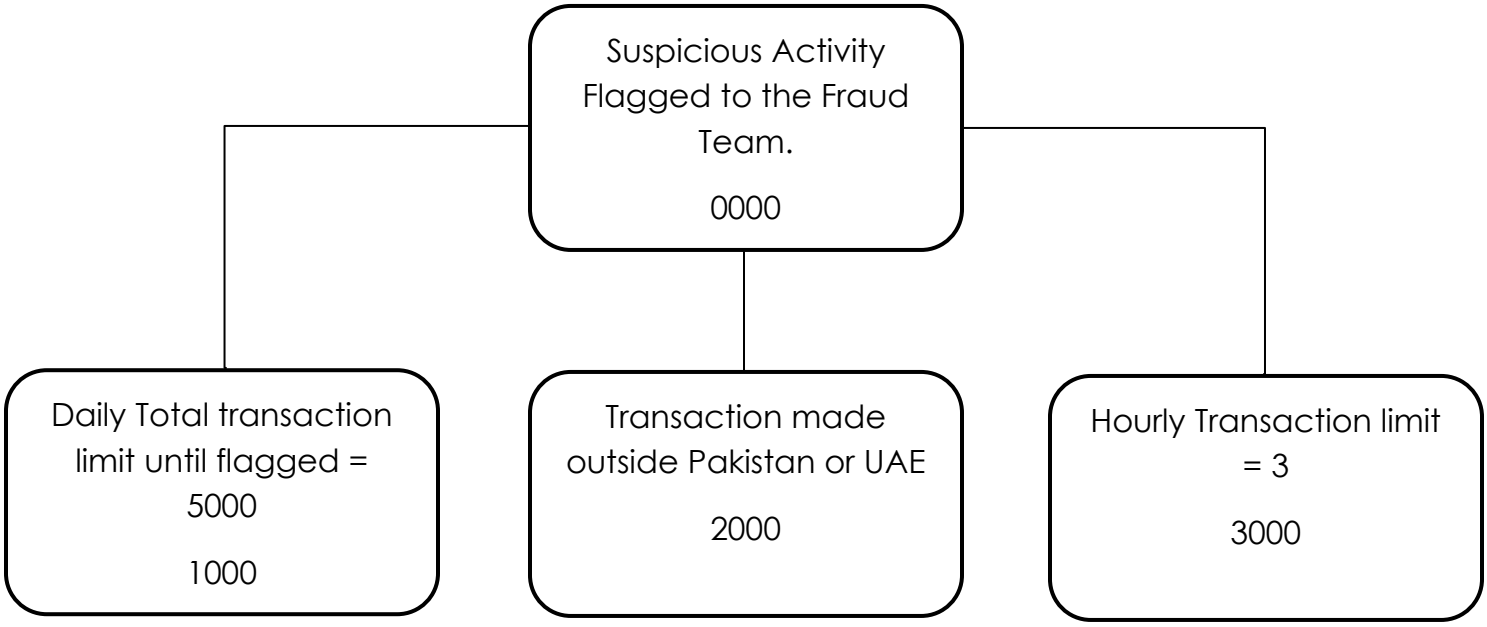
Problem Analysis Chart (PAC):

Given Data	Required Results
• Daily spending limit = 5000 • Transaction in country besides Pakistan and UAE. • No more than 3 transactions per hour.	• Suspicious Activity flagged and reported to the fraud team if any of the conditions are broken.
Processing Required	**Solution Alternatives**
• Daily spending < 5000. • Transaction Location = Pakistan OR UAE. • Transactions in an hour < 4	• Lock the account as soon as any conditions are broken.

Input Processing Output (IPO):

Input	Processing	Module Reference	Output
Daily spending limit = 5000 Transaction = Pakistan OR UAE Transaction < 4	1. Enter previous transaction amount of today. 2. Enter number of transactions made in the last hour. 3. Select country from dropdown list. 4. Enter current transaction amount. 5. Calculate and compare to set conditions. 6. Complete transaction or Report to fraud team. 7. End	Read Read Read Read Calculate Print Transaction control	Transaction completed and/or suspicious Activity reported to fraud team.

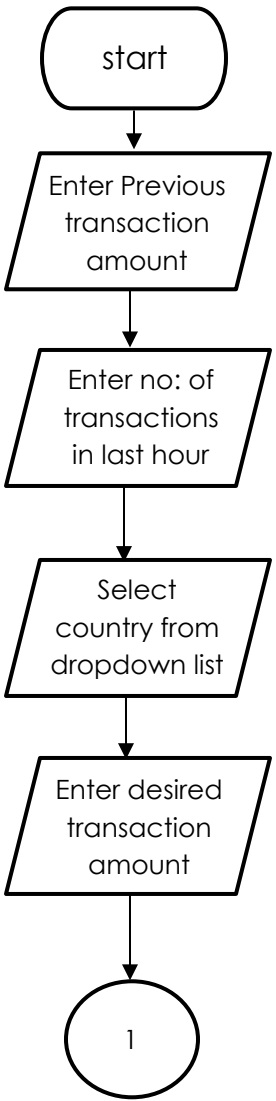
Interactivity Chart (IC):

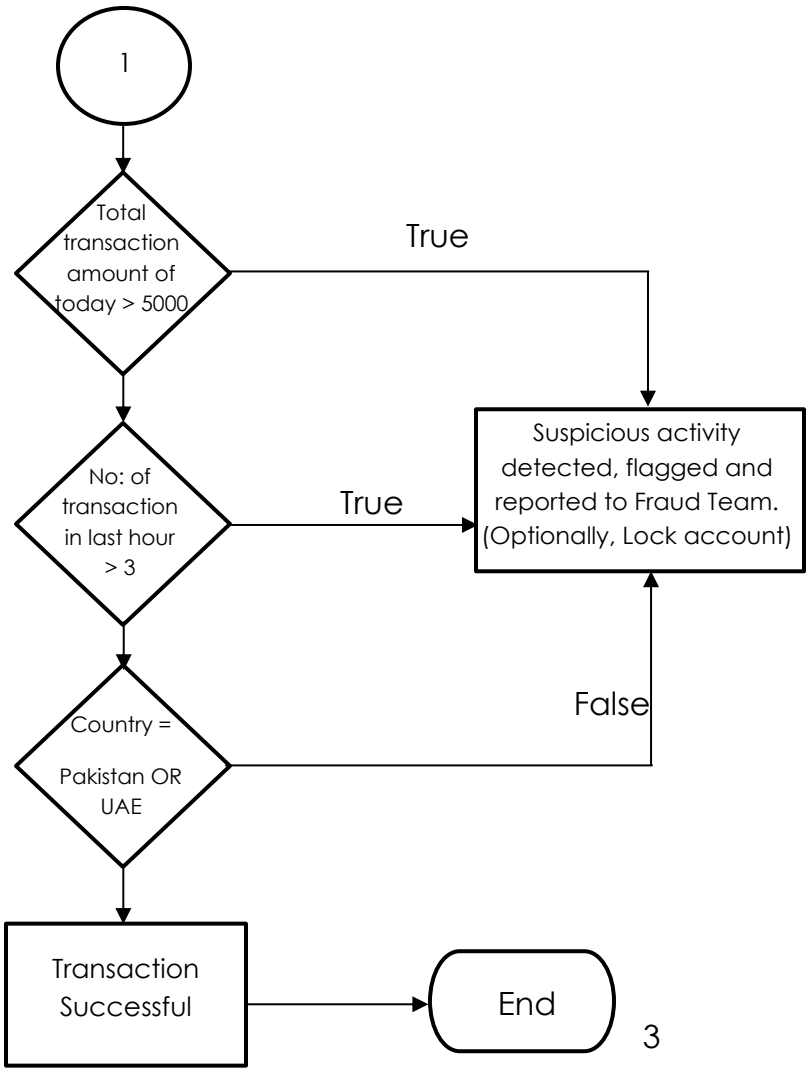


Algorithm:

1. Ask user to enter the transaction amount previously made today.
2. Ask user to enter the number of transactions made in the hour.
3. Ask user to select the country from a dropdown selection list.
4. Ask user for their desired transaction amount.
5. Set (Daily Limit < Previous transaction amount + desired transaction), If true, then flagged as suspicious activity.
6. Set (Transaction in the hour + 1 > 3), if true, then flagged as suspicious activity.
7. Set (Transaction Country = Pakistan OR UAE), if false, then flagged as suspicious activity.
8. If no suspicious activity, Complete transaction.

Flow Chart:





Pseudocode:

```
Print: (Welcome to the bank)
Input: Transaction Amount Today
Input: Transactions Today
Input: Country
Input: Transaction
if (Transaction Amount Today + Transaction > 5000)
                    (Flag activity as suspicious and lock account)
Else if (Transactions Today + 1(current) > 3)
                    (Flag activity as suspicious and lock account)
Else if (Country == Pakistan OR Country == UAE)
                    Print: Transaction successful
Else
(Flag activity as suspicious and lock account).
```

C Code:

```
#include <stdio.h>
int main()
{
   //Declare variables.
int TransactionAmount, TransactionThisHour, DailyTotal;
int Country, Pakistan=17, UAE=35;
/*
Here we will assume the country is being picked from a dropdown selection list.
```

The system will detect and check the numerical placement of the country in the list.

```
*/

   printf("Welcome to the Bank.\n");

   printf("Enter your total previous transaction amount of today.\n"); //previous transactions amount of the day

   scanf("%d", &DailyTotal);

   printf("Enter the number of transactions you have made in this hour.\n"); //number of transactions made this hour

   scanf("%d", &TransactionThisHour);

   printf("Select a country from the list.\n"); //the country where transaction is being made from

   scanf("%d", &Country);

   printf("Enter your desired transaction amount.\n"); //desired transaction amount

   scanf("%d", &TransactionAmount);

  if (DailyTotal+TransactionAmount>5000)

    // checking if the sum of previous transactions and the current transaction exceeds 5000 or not.

  {

   printf("Account access blocked, kindly contact our customer care to unlock it");

    //suspicious transaction flagged for investigation.

  }

  else if (TransactionThisHour + 1>3)

    //checking the transactions made this hour exceed 3 or not.

  {
```

```
    printf("Account access blocked, kindly contact our customer care to unlock it");

     //suspicious transaction flagged for investigation.

   }

   else if (Country==Pakistan || Country==UAE)

     //checking if the transaction is being made at the home country of the client or not, Lets assume Pakistan is 17th on the dropdown list and UAE is 35.

   {

     //Transaction is Normal.

    printf("Transaction successful, Have a nice day.");

   }

   else

   {

    printf("Account access blocked, kindly contact our customer care to unlock it");

     //suspicious transaction flagged for investigation.

   }

   return 0;

}
```

Question 2:

Problem Analysis Chart (PAC):

Given Data	Required Results
• The X and Y component of the point.	• Quadrant Location.
Processing Required	**Solution Alternatives**
• If x>0 and y>0, then Quadrant I. • If x<0 and y>0, then Quadrant II. • If x<0 and y<0, then Quadrant III. • If x>0 and y<0, then Quadrant IV.	• Alongside the Quadrants, the program could recognize if the point is at origin or on a axis.

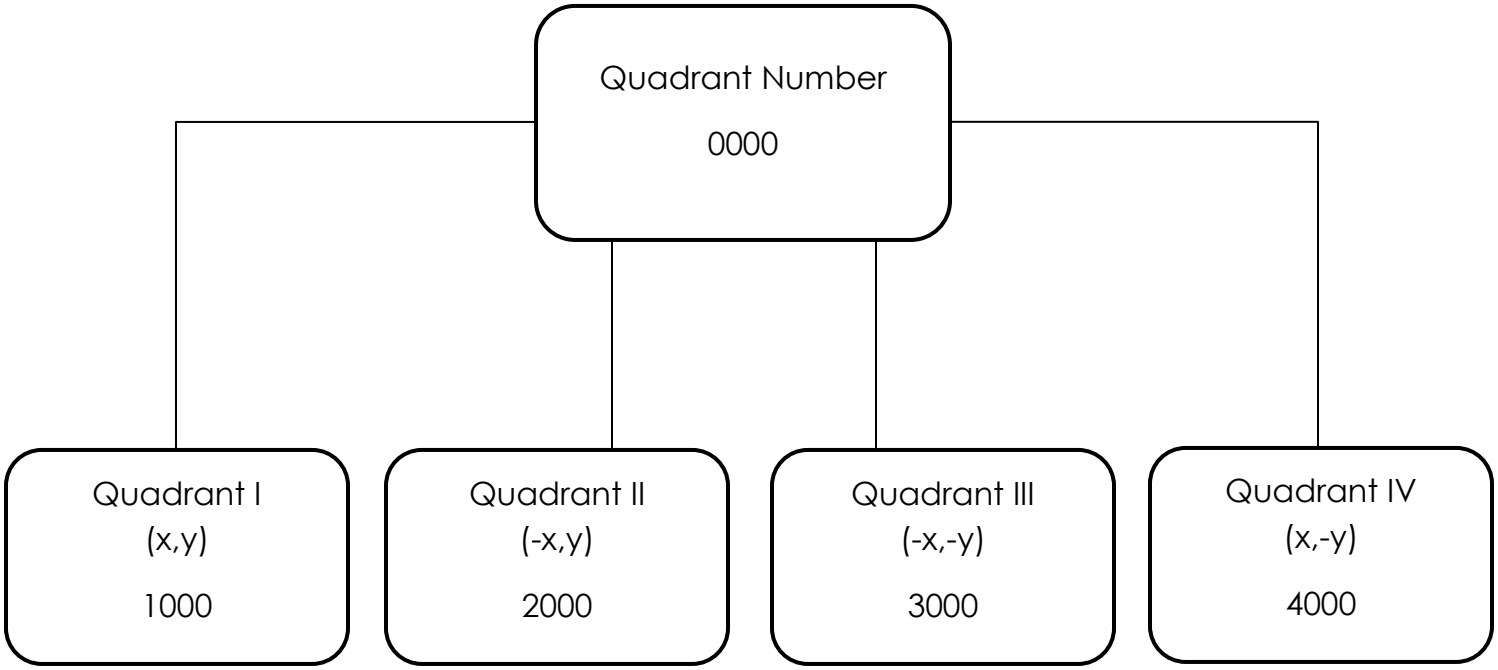
Input Processing Output (IPO):

Input	Processing	Module Reference	Output
The X and Y component of the point.	1. If x>0 and y>0, then Quadrant I. 2. If x<0 and y>0, then Quadrant II. 3. If x<0 and y<0, then Quadrant III. 4. If x>0 and y<0, then Quadrant IV. 5. The point lies in Quadrant (N).	Calculate Calculate Calculate Calculate Print	Quadrant where point lies.

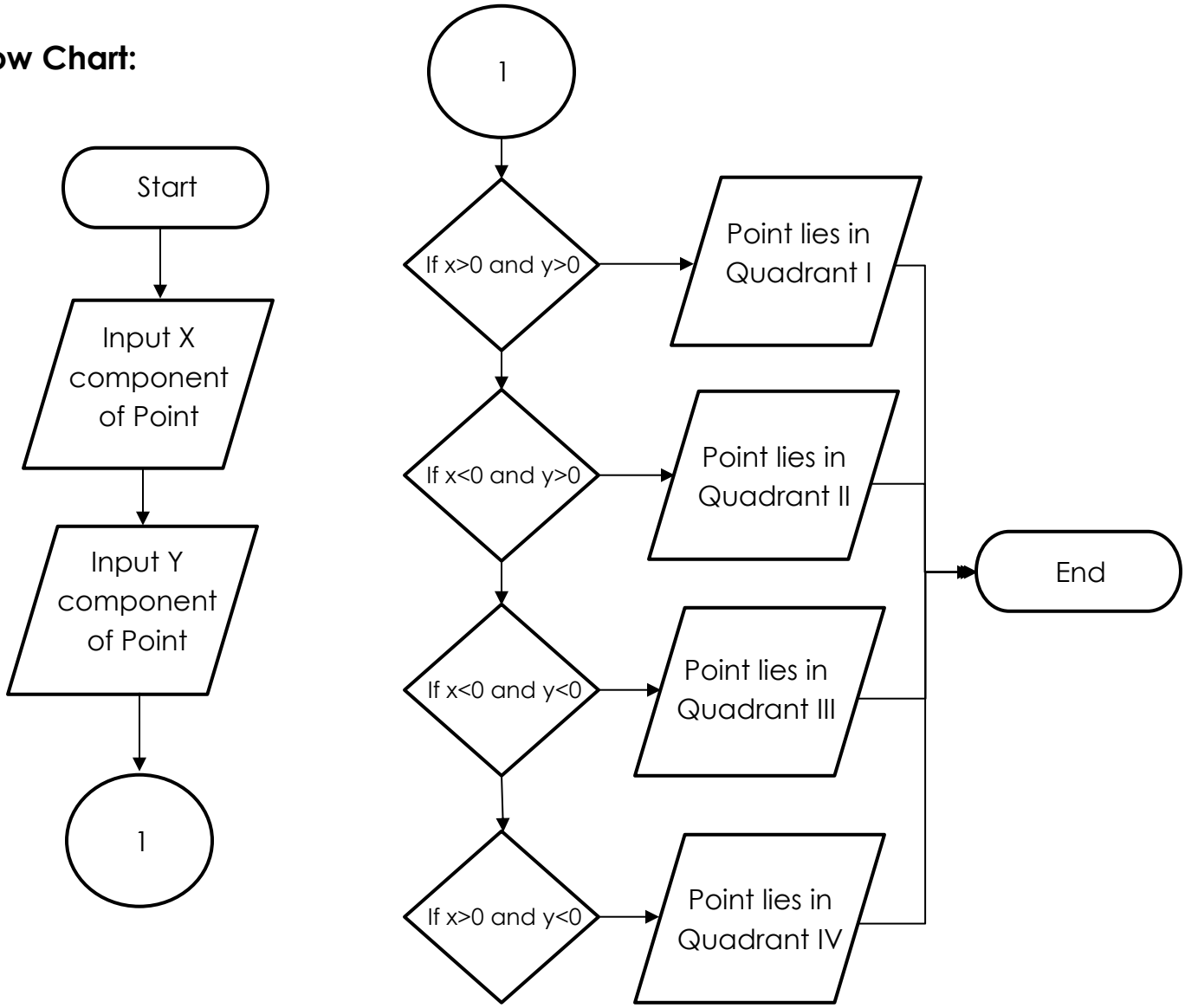
Algorithm:

1. Ask the user to input the x component of the point.
2. Ask the user to input the y component of the point.
3. Set (x > 0 AND Y > 0), Then Quadrant I.
4. Set (x < 0 AND Y > 0), Then Quadrant II.
5. Set (x < 0 AND Y < 0), Then Quadrant III.
6. Set (x > 0 AND Y < 0), Then Quadrant I.
7. Print the Quadrant according to conditions.

Interactivity Chart:



Flow Chart:


Start
Input X component of Point
Input Y component of Point
1
1
If x>0 and y>0
Point lies in Quadrant I
If x<0 and y>0
Point lies in Quadrant II
If x<0 and y<0
Point lies in Quadrant III
If x>0 and y<0
Point lies in Quadrant IV
End

Pseudocode:

```
Input: x component of point
Input: y component of point
If (x>0 and y>0)
        print: The point lies in Quadrant I.
else If (x<0 and y>0)
        print: The point lies in Quadrant II.
Else If (x<0 and y<0)
        print: The point lies in Quadrant III.
Else If (x>0 and y<0)
        print: The point lies in Quadrant IV.
Else If (x=0 and y=0)
        print: The point lies at origin.
Else If (x>0 and y=0)
        print: The point lies in positive x axis.
Else If (x=0 and y>0)
        print: The point lies in positive y axis.
Else If (x<0 and y=0)
        print: The point lies in negative x axis.
Else If (x=0 and y<0)
        print: The point lies in negative y axis.
```

C Code:

```
#include <stdio.h>
int main()
{
    int x,y;
    printf("Insert the point to check the Quadrant.\n");
    printf("Enter the value of x-component: ");
    scanf("%d", &x);
    printf("Enter the value of y-component: ");
    scanf("%d", &y);

    if (x>0 && y>0)
    {
        printf("The point (%d,%d) lies in the 1st Quadrant.",x,y);
    }
    else if (x<0 && y>0)
    {
        printf("The point (%d,%d) lies in the 2nd Quadrant.",x,y);
    }
    else if (x<0 && y<0)
    {
        printf("The point (%d,%d) lies in the 3rd Quadrant.",x,y);
    }
    else if (x>0 && y<0)
    {
        printf("The point (%d,%d) lies in the 4th Quadrant.",x,y);
```

```
    }
    else if (x==0 && y==0)
    {
        printf("The point lies in the Origin.");
    }
    if (x>0 && y==0)
    {
        printf("The point (%d,%d) lies in the positive x-axis.",x,y);
    }
    else if (x==0 && y>0)
    {
        printf("The point (%d,%d) lies in the positive y-axis.",x,y);
    }
    else if (x<0 && y==0)
    {
        printf("The point (%d,%d) lies in the negative x-axis.",x,y);
    }
    else if (x==0 && y<0)
    {
        printf("The point (%d,%d) lies in the negative y-axis.",x,y);
    }

return 0;
}
```

Question 3:

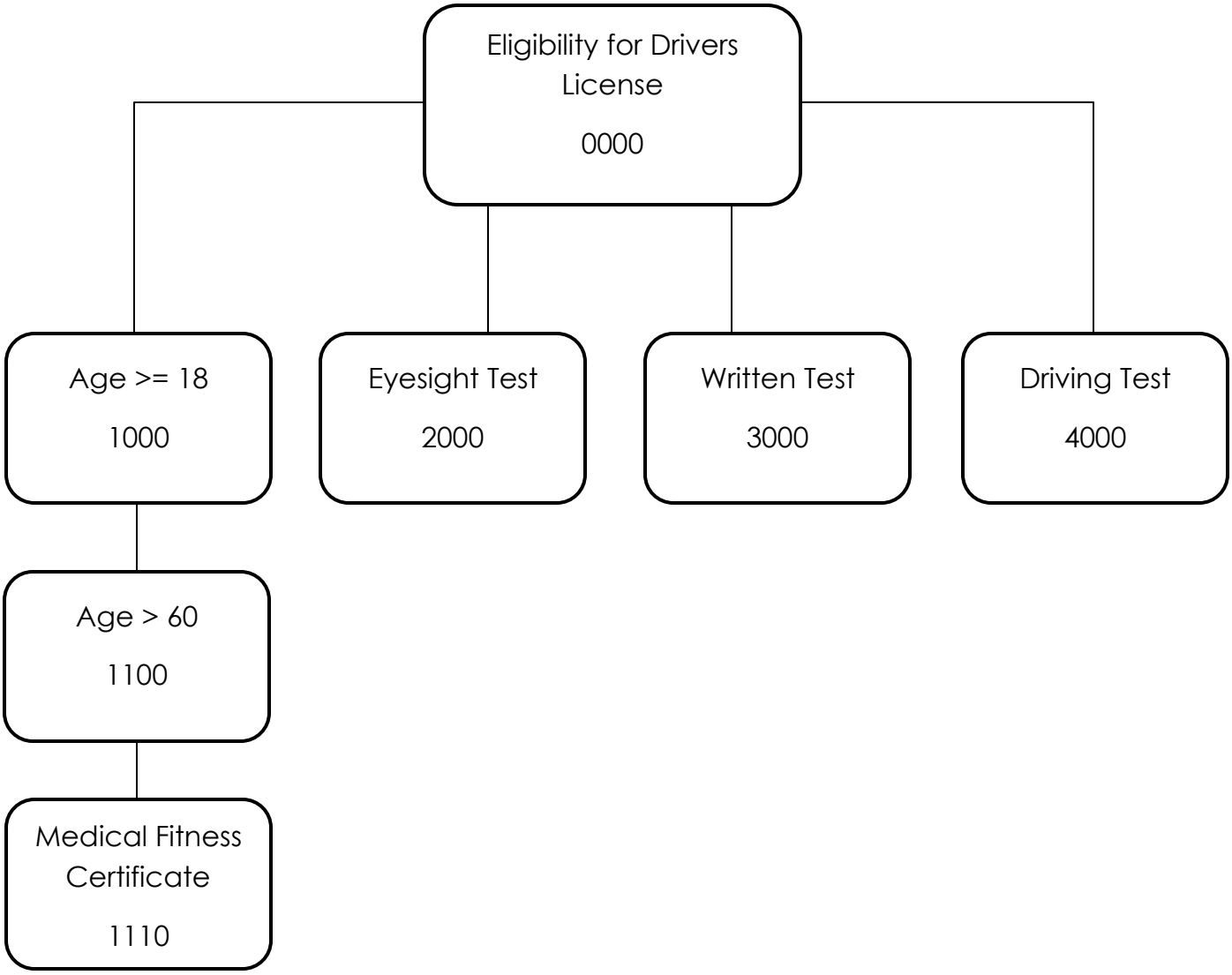
Problem Analysis Chart (PAC):

Given Data	Required Results
• Input Age • Input Eyesight test result. • Input Written test result. • Input Driving test result • Input Medical Fitness Certificate (conditional).	• Eligible for driving license OR • Not eligible for driving license.
Processing Required	**Solution Alternatives**
• Age > 18 • Validate Eyesight Test • Validate Written Test • Validate Driving Test • Validate Conditional Medical Fitness Certificate (If Age > 60)	• Input government identity card. • Input medical checkups + record.

Input Processing Output (IPO):

Input	Processing	Module Reference	Output
1.Input Age 2. Input Eyesight test result. 3. Input Written test result. 4. Input Driving test result 5. Input Medical Fitness Certificate (If Age > 60).	1. Age >= 18 2. Eyesight Test (Pass/Fail) 3. Written Test (Pass/Fail) 4. Driving Test (Pass/Fail) 5. Conditional Medical Fitness Certificate (Yes/No).	Calculate Calculate Calculate Calculate Calculate	Eligible for driving license OR Not eligible for driving license.

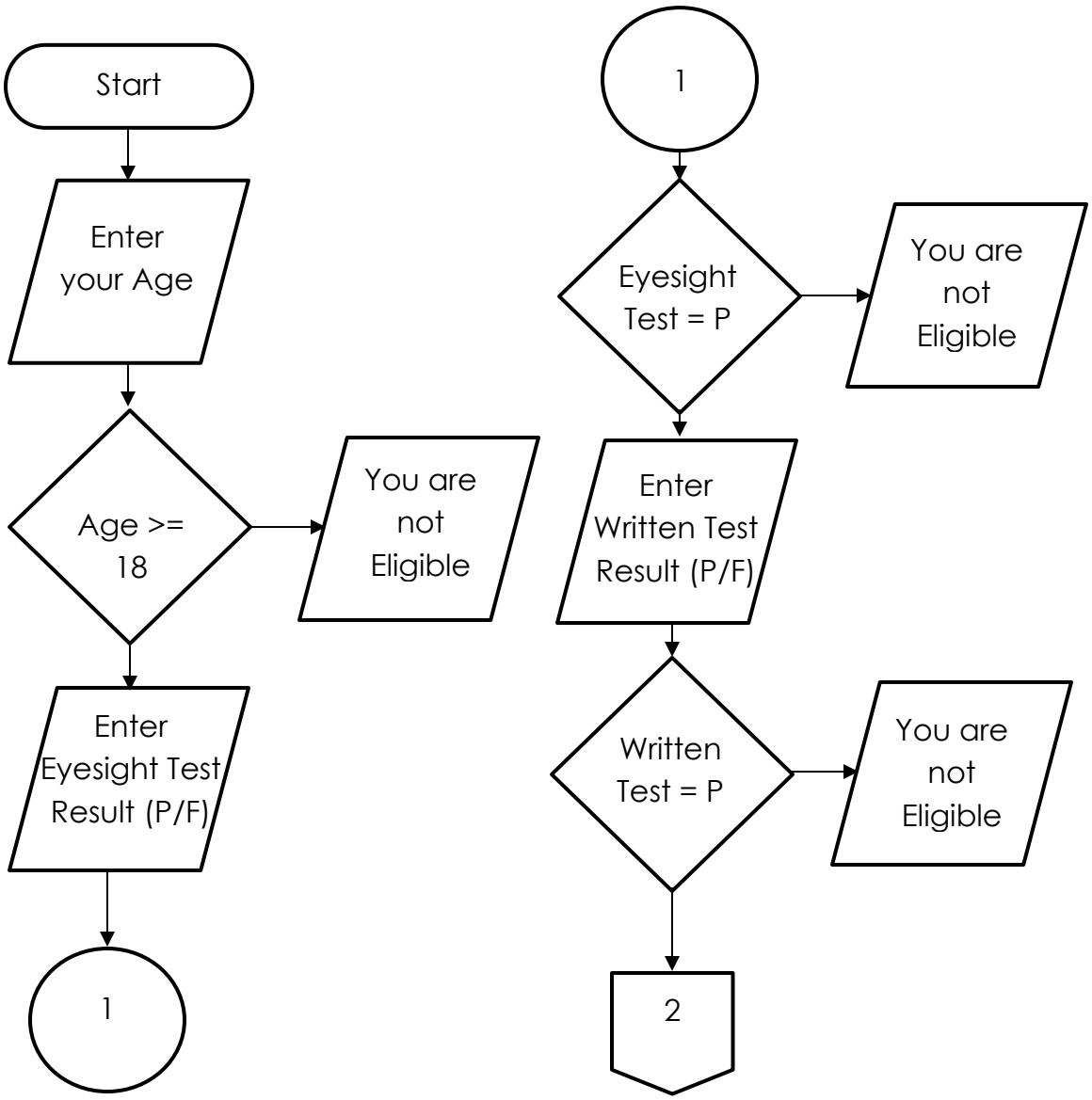
Interactivity Chart:

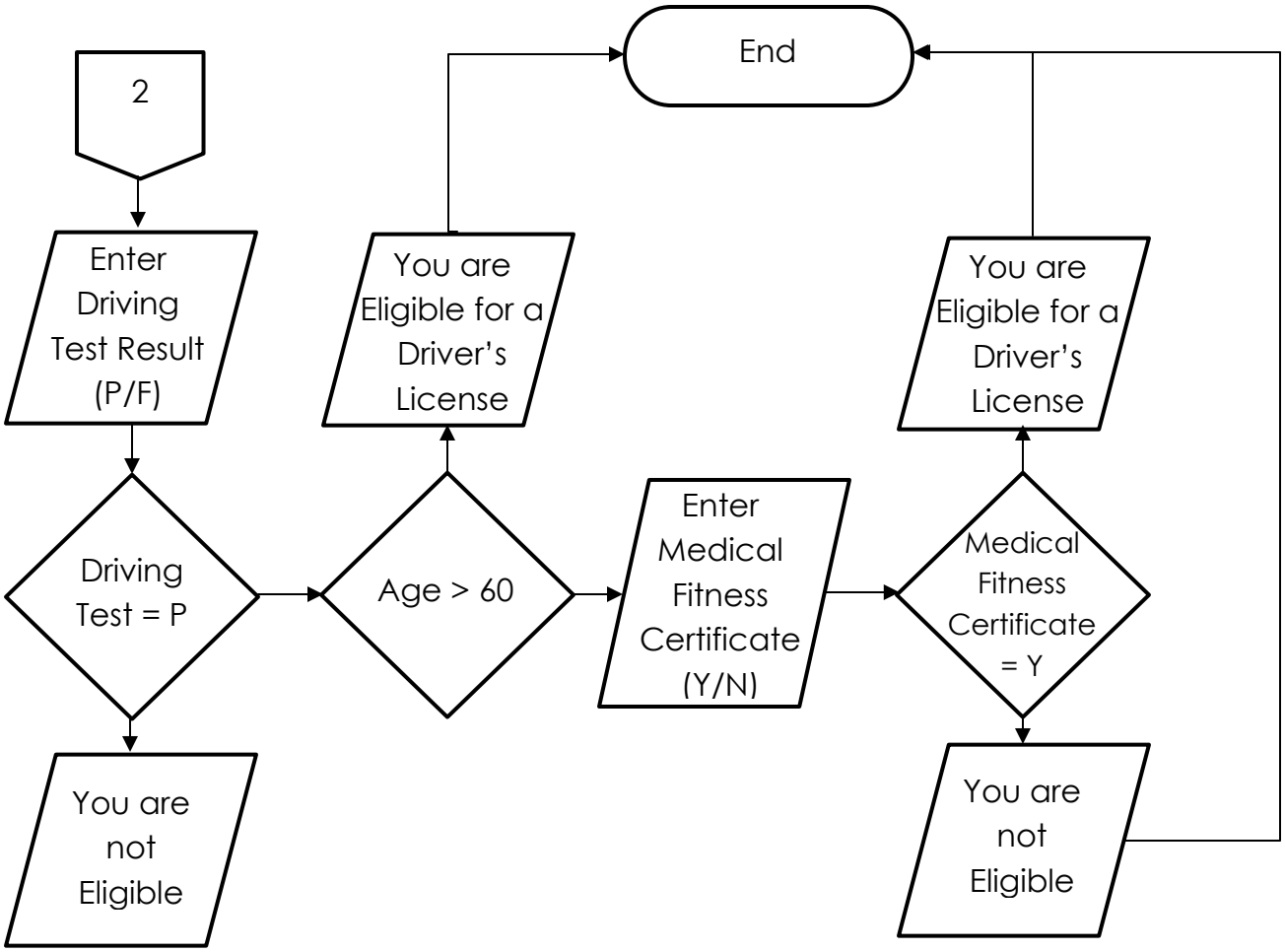


Algorithm:

1. User input Age.
2. If age < 18: You are ineligible, Else: Continue.
3. User input Eyesight Test (Pass/Fail).
4. If Eyesight Test = P: Continue, Else: You are ineligible.
5. User input Written Test (Pass/Fail).
6. If Written Test = P: Continue, Else: You are ineligible.
7. User input Driving Test (Pass/Fail).
8. If Driving Test = P: (if Age<=60: You are eligible, Else: Continue), Else: You are ineligible.
9. User input Medical Fitness Certificate (Yes/No).
10. If Age > 60: Medical Fitness Certificate = Y: You are eligible, Else: You are ineligible.

Flow Chart:





Pseudocode for Version 1:

Input Age

If Age<18: You are not eligible, Else: Continue.

Input Eyesight Test Results (P/F)

If Eyesight Test != P: You are not eligible, Else: Continue

Input Written Test Results (P/F)

If Written Test != P: You are not eligible, Else: Continue

Input Driving Test Results (P/F)

If Driving Test != P: You are not eligible, Else: (If Age < 60: You are eligible, Else: Continue),

Input Medical Fitness Certificate (Y/N)

If Medical Fitness Certificate != Y: You are not eligible, Else: You are Eligible.

C Code Version 1:

```
#include <stdio.h>
int main()
{
   int Age;
   char Eyesight, WrittenTest, DrivingTest, MedicalFitnessCertificate;
   printf("Enter your Age.\n");
   scanf("%d", &Age);
  if (Age<18)
  {
   printf("you are ineligible for a driver's license.");
  }
  else
  {
   printf("Enter P if you passed your Eyesight Test and F if you failed.\n");
   scanf(" %c", &Eyesight);

   if (Eyesight != 'P')
   {
      printf("you are ineligible for a driver's license.");
   }
   else
   {
      printf("Enter P if you passed your Written Test and F if you failed.\n");
      scanf(" %c", &WrittenTest);
```

```
if (WrittenTest != 'P')
{
    printf("you are ineligible for a driver's license.");
}
else
{
    printf("Enter P if you passed your Driving Test and F if you failed.\n");
    scanf(" %c", &DrivingTest);

    if (DrivingTest != 'P')
    {
        printf("you are ineligible for a driver's license.");
    }
    else if (Age>60)
    {
        printf("Enter Y if you have a Medical Fitness Certificate and N if not.\n");
        scanf(" %c", &MedicalFitnessCertificate);

        if (MedicalFitnessCertificate != 'Y')
        {
            printf("you are ineligible for a driver's license.");
        }
        else
        {
            printf("you are Eligible for a driver's license.");
        }
```

```
        }
        else
        {
            printf("you are Eligible for a driver's license.");
        }




      }

    }

   }
return 0;
}
```

Pseudocode for Version 2:

Input Age

If Age>18: Continue, Else: You are not eligible.

Input Eyesight Test Results (P/F)

If Eyesight Test = P: Continue, Else: You are not eligible.

Input Written Test Results (P/F)

If Written Test = P: Continue, Else: You are not eligible.

Input Driving Test Results (P/F)

If Driving Test = P: (If Age > 60: Continue, Else: You are eligible.), Else: You are not eligible.

Input Medical Fitness Certificate (Y/N)

If Medical Fitness Certificate = Y: You are Eligible, Else: You are not Eligible.

C Code Version 2:

```
#include <stdio.h>
int main()
{
   int Age;
   char Eyesight, WrittenTest, DrivingTest, MedicalFitnessCertificate;
   printf("Enter your Age.\n");
   scanf("%d", &Age);
  if (Age>=18)
  {
   printf("Enter your Eyesight Test Results, P for Pass, F for Fail.\n");
   scanf(" %c", &Eyesight);
   if (Eyesight == 'P')
```

```
{
   printf("Enter your Written Test Results, P for Pass, F for Fail.\n");
   scanf(" %c", &WrittenTest);
   if (WrittenTest == 'P')
   {
    printf("Enter your Driving Test Results, P for Pass, F for Fail.\n");
   scanf(" %c", &DrivingTest);
   if (DrivingTest == 'P')
   {
    if (Age>60)
    {
        printf("Enter your Medical Fitness Certificate, Y for Yes, N for No.\n");
        scanf(" %c", &MedicalFitnessCertificate);
        if (MedicalFitnessCertificate == 'Y')
        {
            printf("You are eligible for a Driver's License.");
        }
        else
        {
            printf("You are ineligible for a Driver's license.");
        }

    }
    else
    {
        printf("You are eligible for a Driver's License.");
```

```
     }

    }
    else
    {
     printf("You are ineligible for a Driver's license.");
    }

    }
    else
    {
     printf("You are ineligible for a Driver's license.");
    }

 }
 else
 {
    printf("You are ineligible for a Driver's license.");
 }

}
else
{
 printf("You are ineligible for a Driver's License.");
}
```

```
return 0;
}
```

Question 4:

Problem Analysis Chart (PAC):

Given Data	Required Results
• Input Number1 (let's assume, c1) (1-13) • Input Number2 (let's assume, c2) (1-13) • Input Number3 (let's assume, c3) (1-13) • Input Number4 (let's assume, c4) (1-13) • Input Number5 (let's assume, c5) (1-13)	• It is a House. Or • It is not a House.
Processing Required	**Solution Alternatives**
• Validate c1=c2=c3 AND c4=c5 AND c1!=c4 • Validate c1=c2=c4 AND c3=c5 AND c1!=c3 • Validate c1=c2=c5 AND c3=c4 AND c1!=c3 • Validate c1=c3=c4 AND c2=c5 AND c1!=c2 • Validate c1=c3=c5 AND c2=c4 AND c1!=c2 • Validate c1=c4=c5 AND c2=c3 AND c1!=c2 • Validate c2=c3=c4 AND c1=c5 AND c2!=c1 • Validate c2=c4=c5 AND c1=c3 AND c2!=c1 • Validate c3=c4=c5 AND c1=c2 AND c3!=c1	• A Counter System.

Input Processing Output (IPO):

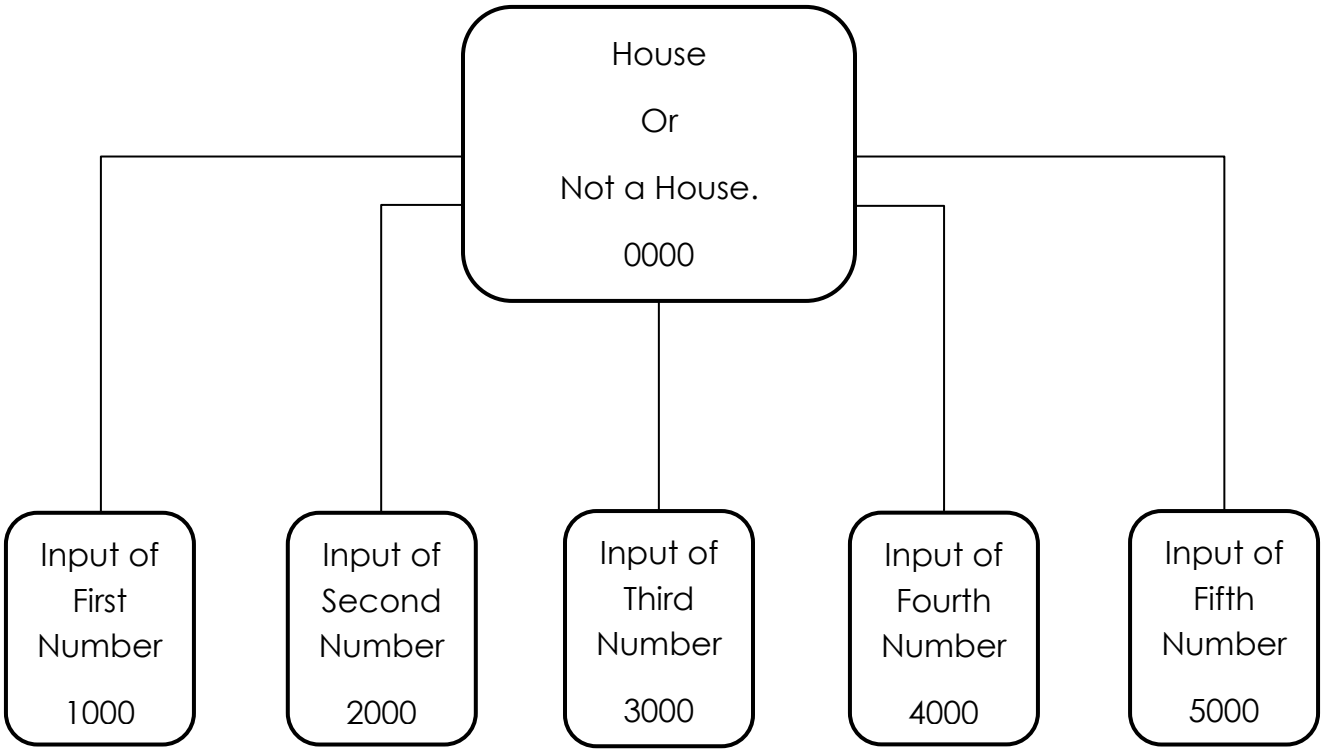
Input	Processing	Module Reference	Output
1.Input Number1 (c1) (1-13) 2.Input Number2 (c2) (1-13) 3.Input Number3 (c3) (1-13) 4.Input Number4 (c4) (1-13) 5.Input Number5 (c5) (1-13)	1. (c1==c2 && c2==c3 && c4==c5 && c1!=c4) 2. (c1==c2 && c2==c4 && c3==c5 && c1!=c3) 3. (c1==c2 && c2==c5 && c3==c4 && c1!=c3) 4. (c1==c3 && c3==c4 && c2==c5 && c1!=c2) 5. (c1==c3 && c3==c5 && c2==c4 && c1!=c2) 6. (c1==c4 && c4==c5 && c2==c3 && c1!=c2) 7. (c2==c3 && c3==c4 && c1==c5 && c2!=c1) 8. (c2==c3 && c3==c5 && c1==c4 && c2!=c1)	Calculate Calculate Calculate Calculate Calculate Calculate Calculate Calculate	It is a House. Or It is not a House.

	9. (c2==c4 && c4==c5 && c1==c3 && c2!=c1) 10. (c3==c4 && c4==c5 && c1==c2 && c3!=c1) 11. It is a House/It is not a House.	Calculate Calculate Print	

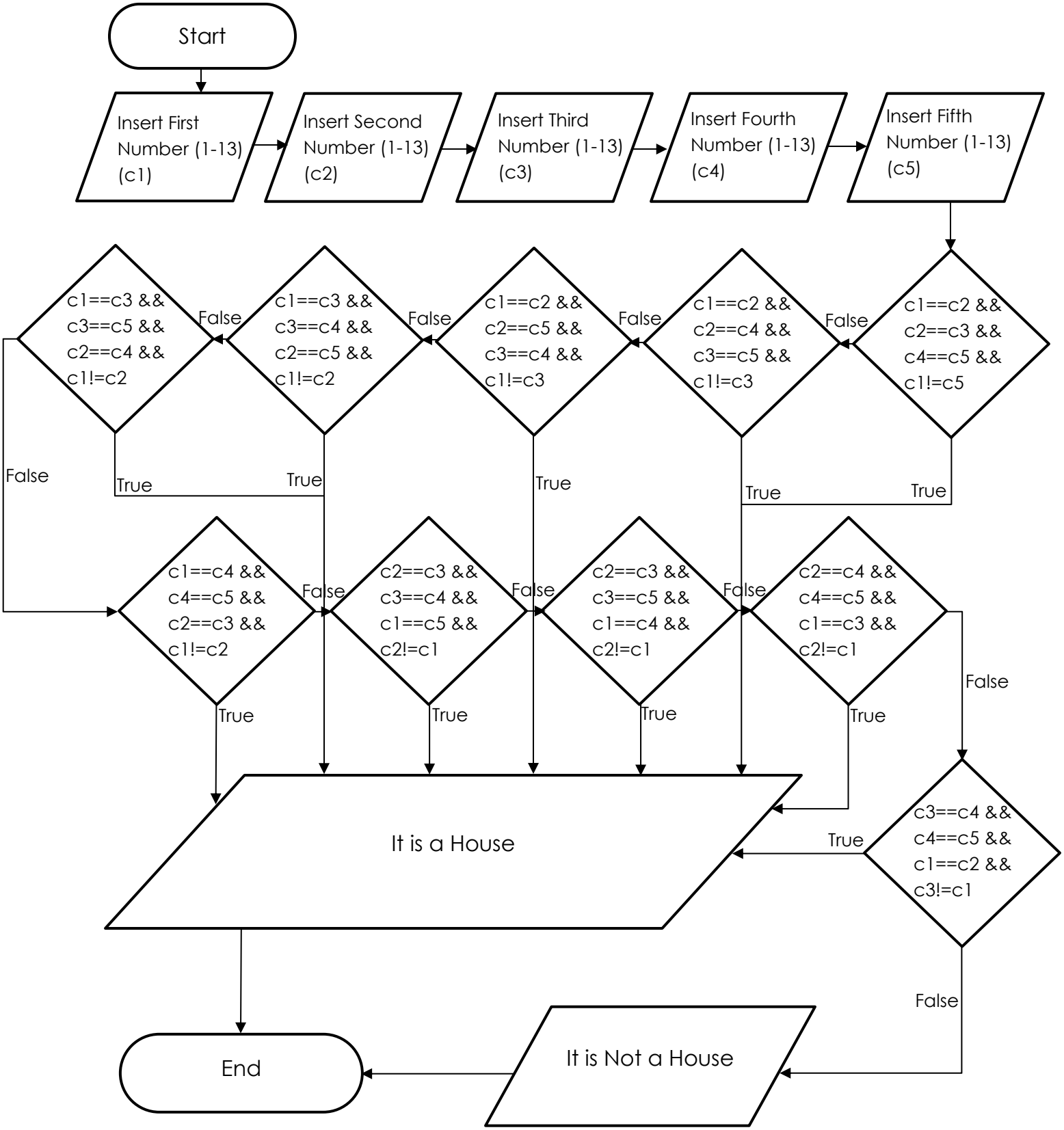
Algorithm:

1. User input Number 1(1-13), assign to c1.
2. User input Number 2(1-13), assign to c2.
3. User input Number 3(1-13), assign to c3.
4. User input Number 4(1-13), assign to c4.
5. User input number 5(1-13), assign to c5.
6. if number is between 1 and 13: continue. Else: Invalid Number.
7. If (c1==c2 && c2==c3 && c4==c5 && c1!=c5): It is a House.
8. Else If (c1==c2 && c2==c4 && c3==c5 && c1!=c3): It is a House.
9. Else If (c1==c2 && c2==c5 && c3==c4 && c1!=c3): It is a House.
10. Else If (c1==c3 && c3==c4 && c2==c5 && c1!=c2): It is a House.
11. Else If (c1==c3 && c3==c5 && c2==c4 && c1!=c2): It is a House.
12. Else If (c1==c4 && c4==c5 && c2==c3 && c1!=c2): It is a House.
13. Else If (c2==c3 && c3==c4 && c1==c5 && c2!=c1): It is a House.
14. Else If (c2==c3 && c3==c5 && c1==c4 && c2!=c1): It is a House.
15. Else If (c2==c4 && c4==c5 && c1==c3 && c2!=c1): It is a House.
16. Else If (c3==c4 && c4==c5 && c1==c2 && c3!=c1): It is a House.
17. Else: It is not a House.

Interactivity Chart:



Flow Chart:


Start
Insert First Number (1-13) (c1)
Insert Second Number (1-13) (c2)
Insert Third Number (1-13) (c3)
Insert Fourth Number (1-13) (c4)
Insert Fifth Number (1-13) (c5)
c1==c2 && c2==c3 && c4==c5 && c1!=c5
c1==c2 && c2==c4 && c3==c5 && c1!=c3
c1==c2 && c2==c5 && c3==c4 && c1!=c3
c1==c3 && c3==c4 && c2==c5 && c1!=c2
c1==c3 && c3==c5 && c2==c4 && c1!=c2
False
True
c1==c4 && c4==c5 && c2==c3 && c1!=c2
c2==c3 && c3==c4 && c1==c5 && c2!=c1
c2==c3 && c3==c5 && c1==c4 && c2!=c1
c2==c4 && c4==c5 && c1==c3 && c2!=c1
c3==c4 && c4==c5 && c1==c2 && c3!=c1
It is a House
It is Not a House
End

Pseudocode:

Input Number 1 (1-13) (c1)

Input Number 2 (1-13) (c2)

Input Number 3 (1-13) (c3)

Input Number 4 (1-13) (c4)

Input Number 5 (1-13) (c5)

If c1, c2, c3, c4, c5 > 0 && c1, c2, c3, c4, c5 <=13: continue, else: Invalid Input.

If (c1==c2 && c2==c3 && c4==c5 && c1!=c5): It is a House, else: continue.

Else if (c1==c2 && c2==c4 && c3==c5 && c1!=c3): It is a House, else: continue.

Else if (c1==c2 && c2==c5 && c3==c4 && c1!=c3): It is a House, else: continue.

Else if (c1==c3 && c3==c4 && c2==c5 && c1!=c2): It is a House, else: continue.

Else if (c1==c3 && c3==c5 && c2==c4 && c1!=c2): It is a House, else: continue.

Else if (c1==c4 && c4==c5 && c2==c3 && c1!=c2): It is a House, else: continue.

Else if (c2==c3 && c3==c4 && c1==c5 && c2!=c1): It is a House, else: continue.

Else if (c2==c3 && c3==c5 && c1==c4 && c2!=c1): It is a House, else: continue.

Else if (c2==c4 && c4==c5 && c1==c3 && c2!=c1): It is a House, else: continue.

Else if (c3==c4 && c4==c5 && c1==c2 && c3!=c1): It is a House, else: continue.

Else: It is not a house.

C Code:

```
#include <stdio.h>
int main()
{
   int c1,c2,c3,c4,c5;
   printf("Enter 1st number from a range of 1 to 13.\n");
   scanf("%d", &c1);
   printf("Enter 2nd number from a range of 1 to 13.\n");
   scanf("%d", &c2);
   printf("Enter 3rd number from a range of 1 to 13.\n");
   scanf("%d", &c3);
   printf("Enter 4th number from a range of 1 to 13.\n");
   scanf("%d", &c4);
   printf("Enter 5th number from a range of 1 to 13.\n");
   scanf("%d", &c5);
   if (0<c1 && c1<=13 && 0<c2 && c2<=13 && 0<c3 && c3<=13 && 0<c4 &&
c4<=13 && 0<c5 && c5<=13)
   {


  if (c1==c2 && c2==c3 && c4==c5 && c1!=c5)
  {
   printf("It is a full House.\n");
  }
  else if (c1==c2 && c2==c4 && c3==c5 && c1!=c3)
  {
   printf("It is a full House.\n");
```

```
}
else if (c1==c2 && c2==c5 && c3==c4 && c1!=c3)
{
 printf("It is a full House.\n");
}
else if (c1==c3 && c3==c4 && c2==c5 && c1!=c2)
{
 printf("It is a full House.\n");
}
else if (c1==c3 && c3==c5 && c2==c4 && c1!=c2)
{
 printf("It is a full House.\n");
}
else if (c1==c4 && c4==c5 && c2==c3 && c1!=c2)
{
 printf("It is a full House.\n");
}
else if (c2==c3 && c3==c4 && c1==c5 && c2!=c1)
{
 printf("It is a full House.\n");
}
else if (c2==c3 && c3==c5 && c1==c4 && c2!=c1)
{
 printf("It is a full House.\n");
}
else if (c2==c4 && c4==c5 && c1==c3 && c2!=c1)
```

```
  {
   printf("It is a full House.\n");
  }
  else if (c3==c4 && c4==c5 && c1==c2 && c3!=c1)
  {
   printf("It is a full House.\n");
  }
  else
  {
   printf("It is not a full house.\n");
  }
   }
   else
   {
       printf("Invalid Number, Please enter numbers between 1 and 13 only.");
   }

return 0;
}
```

Question 5:

Problem Analysis Chart (PAC):

Given Data	Required Results
• Input a number between 0 and 9. • Any number aside from 0 and 9, the program should end.	• The number of occurrences of each number between 0 to 9 shown in a column format.
Processing Required	**Solution Alternatives**
• Add 1 to each number's respective counting variable. • Input of any number aside from 0-9, The program ends.	• Arrays

Input Processing Output (IPO):

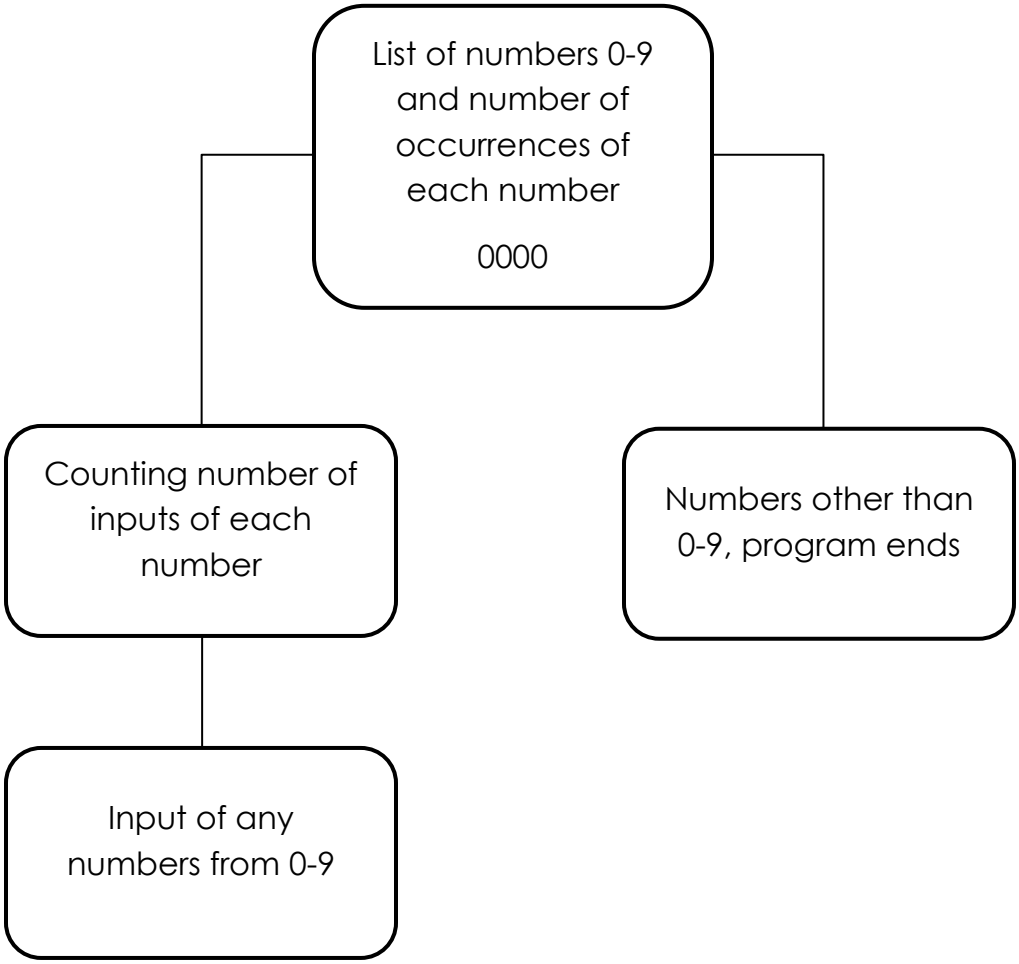
Input	Processing	Module Reference	Output
Input number	1. If the number is 0, Add 1 to its counting variable. 2. If the number is 1, Add 1 to its counting variable. 3. If the number is 2, Add 1 to its counting variable. 4. If the number is 3, Add 1 to its counting variable. 5. If the number is 4, Add 1 to its counting variable. 6. If the number is 5, Add 1 to its counting variable. 7. If the number is 6, Add 1 to its counting variable. 8. If the number is 7, Add 1 to its counting variable. 9. If the number is 8, Add 1 to its counting variable. 10. If the number is 9, Add 1 to its counting variable.	Calculate Calculate Calculate Calculate Calculate Calculate Calculate Calculate Calculate Calculate	The number of occurrences of each number between 0 to 9 being shown in a column format.

	11. If the number is anything else, immediately end the program.	Calculate	

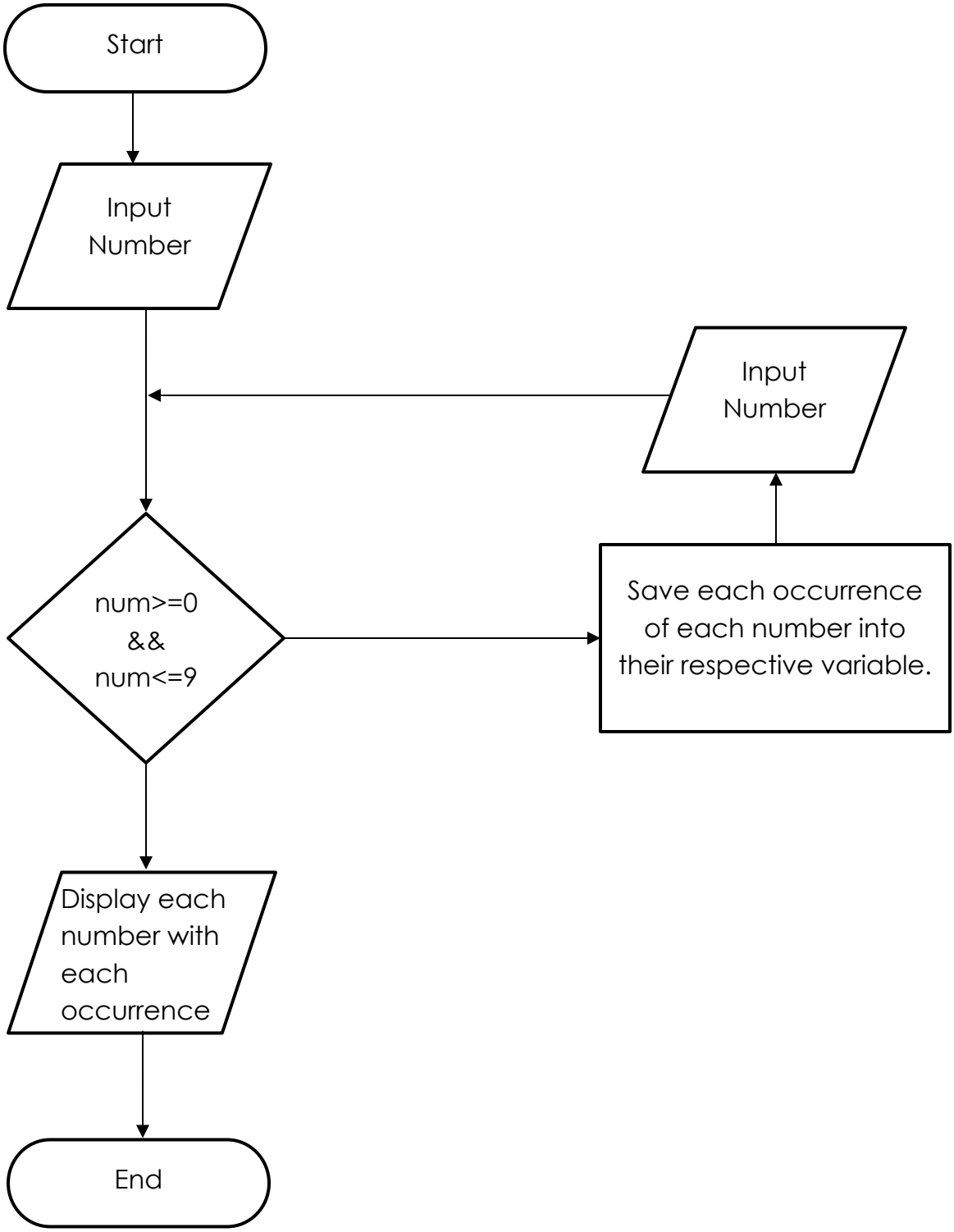
Algorithm:

1. Input any number.
2. Start Loop: [input number num>=0 && num<=9.
3. If input is 0, save to num0 variable.
4. Else If input is 1, save to num1 variable.
5. Else If input is 2, save to num2 variable.
6. Else If input is 3, save to num3 variable.
7. Else If input is 4, save to num4 variable.
8. Else If input is 5, save to num5 variable.
9. Else If input is 6, save to num6 variable.
10. Else If input is 7, save to num7 variable.
11. Else If input is 8, save to num8 variable.
12. Else If input is 9, save to num9 variable.
13. Input any number.]
14. Print each number and variable, in a column format.

Interactivity Chart:



Flow Chart:


Start
Input Number
Input Number
num>=0 && num<=9
Save each occurrence of each number into their respective variable.
Display each number with each occurrence
End

Pseudocode:

```
Input number, save to num.
For(num; num>=0 && num<=9;num)
[If num==0, add 1 to num0.
Else if num==1, add 1 to num1.
Else if num==2, add 1 to num2.
Else if num==3, add 1 to num3.
Else if num==4, add 1 to num4.
Else if num==5, add 1 to num5.
Else if num==6, add 1 to num6.
Else if num==7, add 1 to num7.
Else if num==8, add 1 to num8.
Else if num==9, add 1 to num9.
Input number.]
Print(number      number of occurrences)
Print(    0                 num0)
Print(    1                 num1)
Print(    2                 num2)
Print(    3                 num3)
Print(    4                 num4)
Print(    5                 num5)
Print(    6                 num6)
Print(    7                 num7)
Print(    8                 num8)
Print(    9                 num9)
```

C Code:

```
#include <stdio.h>

int main()

{

   int num, num0=0, num1=0, num2=0, num3=0, num4=0, num5=0, num6=0,
num7=0, num8=0, num9=0;

   printf("Enter a number between 1 and 9.\n");

   scanf("%d", &num);

      for (num;num>=0 && num<=9;num)

      {

         if (num==0)

         {

            num0++;

         }

         else if (num==1)

         {

            num1++;

         }

         else if (num==2)

         {

            num2++;

         }

         else if (num==3)

         {

            num3++;

         }
```

```
 else if (num==4)
{
    num4++;
}
 else if (num==5)
{
    num5++;
}
 else if (num==6)
{
    num6++;
}
else if (num==7)
{
    num7++;
}
else if (num==8)
{
    num8++;
}
else if (num==9)
{
    num9++;
}
scanf("%d",&num);
```

```
        }
    printf("\n\n\n");
    printf("Number \t\t Number of Occurrences\n");
    printf("  0 \t\t\t   %d \n", num0);
    printf("  1 \t\t\t   %d \n", num1);
    printf("  2 \t\t\t   %d \n", num2);
    printf("  3 \t\t\t   %d \n", num3);
    printf("  4 \t\t\t   %d \n", num4);
    printf("  5 \t\t\t   %d \n", num5);
    printf("  6 \t\t\t   %d \n", num6);
    printf("  7 \t\t\t   %d \n", num7);
    printf("  8 \t\t\t   %d \n", num8);
    printf("  9 \t\t\t   %d \n", num9);
return 0;
}
```

Question 6:

Consider the following section of C code:

```
// i, j, and k are ints
if (i < j)
{
if (j < k)
i = j;
else
j = k;

}
else
{
if (j > k)
j = i;
else
i = k;

}
printf("%d %d %d\n", i, j, k);
```

What will the code print if the variables i, j, and k have the following values?

(a) i is 3, j is 5, and k is 7

Start: i=3, j=5, k=7

Check: i<j --- 3<5 (True)

Check: j<k --- 5<7 (True)

Assigning: i=j --- i=5 (j=5)

Print: 5 5 7.

(b) i is 3, j is 7, and k is 5

Start: i=3, j=7, k=5

Check: i<j →3<7 (True)

Check: j<k → 7<5 (False)

Assigning: j=k → j=5 (k=5)

Print: 3 5 5.

(c) i is 5, j is 3, and k is 7

Start: i=5, j=3, k=7

Check: i<j →5<3 (False)

Check: j>k → 3>7 (False)

Assigning: i=k → i=7 (k=7)

Print: 7 3 7.

(d) i is 5, j is 7, and k is 3

Start: i=5, j=7, k=3

Check: i<j →5<7 (True)

Check: j<k → 7<3 (False)

Assigning: j=k → j=3 (k=3)

Print: 5 3 3.

(e) i is 7, j is 3, and k is 5

Start: i=7, j=3, k=5

Check: i<j →7<3 (False)

Check: j>k → 3>5 (False)

Assigning: i=k → i=5 (k=5)

Print: 5 3 5.

(f) i is 7, j is 5, and k is 3

Start: i=7, j=5, k=3

Check: i<j →7<5 (False)

Check: j>k → 5>3 (True)

Assigning: j=i → j=7 (i=7)

Print: 7 7 3.

Question 7:

Part A: Corrected code.

```
#include <stdio.h>
int main()
{
int value;
printf("Please enter a value in the range of 1 to 7: ");
scanf("%d", &value);

if (value == 1)
{
 printf("The day is Monday.");
}
else if (value == 2)
{
 printf("The day is Tuesday.");
}
else if (value == 3)
{
 printf("The day is Wednesday.");
}
else if (value == 4)
{
 printf("The day is Thrusday.");
}
```

```
else if (value == 5)
{
 printf("The day is Friday.");
}
else if (value == 6)
{
 printf("The day is Saturday.");
}
else if (value == 7)
{
 printf("The day is Sunday.");
}
else
{
 printf("You entered a value out of range.");
}
return 0;
}
```

Part B: Rewritten Code using Switch-Case Statements.

```
#include <stdio.h>

int main()

{

   int Day;

   printf("Enter a value between 1-7 and I will state the corresponding day of the week.\n");

   scanf("%d", &Day);

   switch (Day)

   {

   case 1:

      printf("The day is Monday.");

      break;

   case 2:

      printf("The day is Tuesday.");

      break;

   case 3:

      printf("The day is Wednesday.");

      break;

   case 4:

      printf("The day is Thursday.");

      break;

   case 5:

      printf("The day is Friday.");

      break;

   case 6:
```

```
        printf("The day is Saturday.");
        break;
    case 7:
        printf("The day is Sunday.");
        break;
    default:
        printf("The input value is out of range of 1-7.");
        break;
    }
return 0;
}
```